

COURSE TITLE : METALLURGY AND MATERIALS OF CONSTRUCTION
COURSE CODE : 5074
COURSE CATEGORY : E
PERIOD/WEEK : 4
PERIOD/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to Metallurgy.	13
2	Know materials of metallic construction and its applications	13
3	Know materials of Non- metallic construction and its applications	13
4	Failure of materials of construction due to corrosion and its prevention.	13
TOTAL		52

COURSE OBJECTIVE :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Understand principles of metallurgy. Comprehend the extraction of metal from its ores.
	2	Know materials of metallic construction and its applications
	3	Know materials of Non- metallic construction and its applications
2	4	Understand the failure of materials of construction due to corrosion and its prevention..

SPECIFIC OUTCOMES

MODULE I

1.1.0 UNDERSTAND THE PRINCIPLES OF GENERAL METTULLURGY

- 1.1.1 List the physical properties of ores.
- 1.1.2 List the various steps involved in the extraction of metals from their ores
- 1.1.3 Explain the working principles of gravity separation processes
- 1.1.4 Explain the magnetic separation process with a neat sketch
- 1.1.5 Explain the principles of froath flotation process with a neats ketch
- 1.1.6 Draw and explain the working of at least two floatation cells used in the industry
- 1.1.7 Explain the working principle of ore concentration by roasting and calcinations
- 1.1.8 List the twenty four characteristics of metals
- 1.1.9 Explain the methods of purification of extracted metals

1.2.0 UNDERSTAND THE METHOD OF EXTRACTION OF IRON FROM ITS ORE

- 1.2.1 List the ores of iron
- 1.2.2 Explain the various steps involved in the concentration of iron ore
- 1.2.3 Explain the method of extraction of iron using blast furnace
- 1.2.4 Draw the sketch of a blast furnace and name the parts and show the temperature
- 1.2.5 Explain the steps involved in the conversion of pig iron to cast iron
- 1.2.6 List the six varieties of cast iron
- 1.2.7 List the physical and chemical properties of cast iron

1.3.0 UNDERSTAND THE CONVERSION OF IRON TO STEEL AND THE PROPERTIES OF STEEL

- 1.3.1 Explain the steps involved in the manufacture of steel by open hearth process
- 1.3.2 Draw a neat sketch of open hearth furnace and name the parts

1.4.0 UNDERSTAND THE PRINCIPLES OF EXTRACTION OF COPPER FROM THEIR ORES

- 1.4.1 List the five ores of copper
- 1.4.2 Explain the various steps involved in the extraction of copper from copper pyrites with the help of chemical equations
- 1.4.3 Name the physical and chemical properties of copper
- 1.4.4 List the uses of copper

1.5.0 UNDERSTAND THE PRINCIPLE OF EXTRACTION OF ALUMINIUM FROM ITS ORES

- 1.5.1 List the four ores of Aluminum
- 1.5.2 Explain the steps involved in the concentration of bauxite with the help of chemical equation
- 1.5.3 Draw a neat sketch of electrolytic refining cell
- 1.5.4 Explain purification of aluminum by Hoopes process
- 1.5.5 Name the physical and chemical properties of aluminum
- 1.5.6 List the uses of aluminum

1.6.0 UNDERSTAND THE PRINCIPLE OF EXTRACTION OF ZINC FROM THEIR ORES

- 1.6.1 Name the two ores of zinc
- 1.6.2 Explain the various steps involved in the extraction of zinc from zinc blend using chemical equation
- 1.6.3 Draw a neat sketch of the equipment used in refining of zinc and name its parts
- 1.6.4 Name the physical and chemical properties of zinc
- 1.6.5 List the uses of zinc

1.7.0 UNDERSTAND THE PRINCIPLE OF EXTRACTION OF TITANIUM FROM THEIR ORES

- 1.7.1 Name the two ores of Titanium
- 1.7.2 Explain the various steps involved in the extraction of Titanium from ilmenite with the help of chemical equation
- 1.7.3 Name the physical and chemical properties of Titanium

MODULE II

2.1.0 UNDERSTAND THE PROPERTIES, COMPOSITION AND APPLICATION OF THE METTALIC MATERIALS OF CONSTRUCTION

- 2.1.1 List the composition of the following metallic materials of construction
Cast Iron, Mild steel, Nickel steel, Chrome Steel, Silicon Steel
- 2.1.2 List the physical properties of the above metallic materials of construction
- 2.1.3 List the applications of above metallic materials of construction with specific industrial examples
- 2.1.4 Classify the various grades of Stainless steel
- 2.1.5 Explain the effect of impurity and causes for failure Steels
- 2.1.6 List the physical properties and uses of various grades of stainless Steel
- 2.1.7 List the application of lead in chemical industries showing specific example
- 2.1.8 List the application of Titaniumin chemical industries showing specific example

MODULE III

3.1.0 UNDERSTAND THE PROPERTIES AND USES OF NON METALLIC MATERIALS OF CONSTRUCTION USED IN THE CHEMICAL INDUSTRY

- 3.1.1 Explain the properties and application of the following with reference to chemical industries 1.Glass 2.ceramics 3. Carbon and refractories
- 3.1.2 List the specific application of metallic and Non metallic materials in the following industries 1) Sulphuric acid 2) Nitric acid 3) Acetic acid 4) Chloralkali 5) Hydrochloric acid 6) Phosphoric acid 7) Ammonia 8) Petroleum 9) Food processing
- 3.1.3 List the application of the following in chemical industries a) Claddings b) lining and coatings of lead c)tin d)rubber e)plastic
- 3.1.4 Explain their Properties Claddings, lining and coatings of lead, tin, rubber and plastic

MODULE IV

4.1.0 UNDERSTAND THE FAILURE OF MATERIALS OF CONSTRUCTION DUE TO CORROSION AND ITS PREVENTION

- 4.1.1 Classify the corrosion reactions and explain the mechanism of corrosion
- 4.1.2 Explain the chemistry of corrosion reactions and explain the mechanism of corrosion
- 4.1.3 Distinguish between the anodic and cathodic reactions
- 4.1.4 List the galvanic series
- 4.1.5 Distinguish between the corrosion reactions in homogeneous and heterogeneous alloys
- 4.1.6 List the factors affecting the corrosion reaction
- 4.1.7 List and explain the six methods of reducing corrosion rate
- 4.1.8 List and explain the eight more frequently encountered forms of corrosion.

CONTENT DETAILS

MODULE I

General principle of Ferrous and Non ferrous metallurgy

Properties of ores – ore beneficiation by gravity separation, magnetic separation, froath floatation, calcinations and roasting

Extraction of pig iron from ore – Blast furnace. Manufacture of steel by Bessemer and open hearth process

Extraction of the following metals 1) Copper 2) Aluminum 3) Zinc 4) Titanium

MODULE II

MATERIALS OF CONSTRUCTION - METALLIC

Properties of composition and application of the following materials with specific examples

a) Cast Iron, Mild steel

b) Alloy Steel – Nickel Steel, Chrome Steel, Chrome Molybdenum Steel, Silicon steel, Manganese Steel, Duriron

c) Stainless Steel - Classification-Variou grades and their standards ISI and AISI.

d) Copper alloys– brass, Bronze, Phosphor Bronze, Cupronickels

e) Nickel alloys–Hastelloy, monel, Inconel, Alloy20.

f) Aluminum alloys

g) Tantalum and Titanium

h) Lead

MODULE III

MATERIALS OF CONSTRUCTION - NON METALLIC

Properties and application of glass, Ceramics, Carbon and Refractories

Specific application of metallic and non metallic materials in the following industry

a) Sulphuric acid, b) Hydrochloric acid c) Nitric Acid, d) Phosphoric acid e) Acetic acid

f) Ammonia g) Petroleum h) Food processing i) Chloralkali j) Food processing -Cladding, Lining and coatings of Lead, Tin, Glass, Rubber and Plastics– their applications

MODULE IV

CORROSION-

Fluid Corrosion – Metallic and Non metallic localized pitting Galvanic corrosion – reaction, anodic, cathodic, galvanic series Inter granular corrosion – Stress corrosion-Cracking liquid metal corrosion-fatigue Cavitation- Fretting corrosion– hydrogen attack – fluid corrosion – Structural, graphitic, pitting. De alloying and De zincification – Biological corrosion – Factors influencing corrosion-Solution

pH, Oxidizing agents, Temperature, velocity, films and other effects.

High temperature attack

Combating corrosion – material selection, proper de altering the environment, inhibitor, cathodic protection, anodic protection, coatings, and linings.

REFERENCE

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|-----------------|---|--------------------------------------|
| O.P.Ghanna | - | Material Science and Metallurgy |
| L.Krishna Reddy | - | Principles of Engineering Metallurgy |